New Zealand Residential EV Charger Installation Guide

32 A single-phase Mode 3 (7.4 kW) wall-mounted EVSE
Standards, regulatory requirements, worked examples, and installer checklist
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Disclaimer: This document is an educational summary for homeowners, designers, and electrical workers. It is not legal advice and does not replace the full text of cited standards, regulations, or network rules. Always use the current editions of AS/NZS 3000 (with applicable amendments), the Electricity (Safety) Regulations 2010, WorkSafe guidance, and your electricity distributor's requirements. Confirm transition dates and local rules with a licensed electrical worker and, where needed, an electrical inspector.

Contents

1. Regulatory framework

Residential EV charger installation in New Zealand is governed by a layered framework:

Layer	Instrument	What it controls
Act	Electricity Act 1992	General electrical safety duties

Regulations	Electricity (Safety) Regulations 2010 (ESR)	Installation rules, product conformity, certification
Cited standard	Schedule 2 ESR → AS/NZS 3000	Wiring Rules: design, construction, verification
Regulation 59	ESR Reg 59	All low-voltage installations must comply with Part 2 of AS/NZS 3000
Guidance	WorkSafe EV Charging Safety Guidelines (3rd ed.) + Addendum (Jan 2026)	Industry/regulator expectations for EVSE
Licensing	Electrical Workers Registration Board (EWRB)	Only registered electrical workers may carry out prescribed electrical work
Network	Distributor Service & Installation Rules	Load limits, notification, controlled loads (varies by region)

Key point: Only a **registered electrical worker** may install a fixed Mode 3 EV charger. The installer must issue a **Certificate of Compliance (CoC)** for the work.

2. Which standards and amendments to use

2.1 Legally cited Wiring Rules (from 13 November 2025)

The Electricity (Safety) Amendment Regulations 2025 (SL 2025/225) updated Schedule 2 of the ESR:

Previously cited	Now cited (current)
AS/NZS 3000:2007 (Amendments 1 & 2) with NZ modifications	**AS/NZS 3000:2018 including Amendments 1, 2, and 3**

2.2 Amendment summary

Amendment	Published	Relevance to residential 32 A EV
Amendment 1 (2020)	Jan 2020	**Clause 7.9 (NZ ONLY)** — mandatory EV charging rules for NZ
Amendment 2 (2021)	Apr 2021	RCD on alterations; switchboard replacement; polarity/MEN testing for outbuildings
Amendment 3 (2023)	May 2023	Switchboard/switchgear standard references (not new 32 A EV rules)

2.3 Companion standards

Standard	Use
AS/NZS 3008.1.1	Cable current-carrying capacity and voltage drop

IEC 61851-1 / 61851-22	Mode 3 EVSE product safety
IEC 62955	Residual Direct Current Detecting Device (RDC-DD)
IEC 62752	In-cable control & protection device (IC-CPD) if plug-in Mode 2
IEC 62196	EV connectors (cited in ESR Schedule 2)
AS/NZS 60335.2.29	Battery chargers (ESR Schedule 4, updated Nov 2025)

3. Transition timeline (2025 regulatory update)

Date / situation	Requirement
13 November 2025 onward	May choose to work to AS/NZS 3000:2018 + Amd 1, 2, 3
Alteration started after 13 Nov 2025, not complete by 12 Nov 2026	CoC for work done; remainder must meet 2018 + amendments
New installation construction starts after 12 November 2026	**Must** comply with AS/NZS 3000:2018 + Amd 1, 2, 3
Repairs (like-for-like, same location)	No whole-installation upgrade required if not unsafe

Example — timing: A homeowner books a 32 A wallbox install in March 2026. The electrician should design and certify to **AS/NZS 3000:2018 + all three amendments** and Clause **7.9.3.3**, even during the transition window, to avoid rework after November 2026.

4. NZ mandatory requirements — Clause 7.9.3.3 (Mode 3 & 4, domestic)

Clause 7.9 applies in New Zealand only. For a residential Mode 3 AC charger (~7.4 kW, 32 A single-phase), Clause **7.9.3.3** requires:

Item	Requirement (7.9.3.3)
(a) Subcircuit capacity	Single-phase final subcircuit minimum **32 A** current-carrying capacity
(b) Dedicated circuit	Final subcircuit shall **not supply any other point** in the wiring
(c) RCD	Separate **Type B** RCD, 30 mA max, operates in **all live conductors** (active and neutral). RCBO permitted
(c) Exception	**Type A** RCD 30 mA + **RDC-DD** conforming to IEC 62955 (≥ 6 mA DC detected)
(d) Connection	**Direct connection** (fixed wiring) per Clause 4.3.2.1 — not a shared general socket circuit
(e) Isolator	Isolating switch ≥ **32 A**, adjacent to charging facility, per Clause 2.3.2.2.1
(f) Height	Socket-outlet or EVSE cable installed ≥ **800 mm** above floor/ground
NOTE	If equipment > 20 A single-phase: consider upgrading mains and/or **load management**

Also: Clause 7.9.2.2 — socket-outlets shall **not** be installed for **Mode 1** charging.

New domestic installations with garage: Clause 7.9.3.1 **requires** a facility for electric vehicle charging in every new domestic installation that has a garage (fully enclosed space with a door for a motor vehicle, incorporated in the main dwelling). That facility must comply with either Clause 7.9.3.2 (Mode 2) or Clause 7.9.3.3 (Mode 3/4).

5. WorkSafe EV guidelines and January 2026 Addendum

WorkSafe's *Electric vehicle charging safety guidelines* (3rd edition) plus the **January 2026 Addendum** should be read alongside AS/NZS 3000. Highlights for **residential EVSE**:

Topic	Addendum / guideline position
RCD	Residential charging final subcircuit: **Type B RCD**
Origin	Subcircuit shall originate from a **MEN switchboard**
RDC-DD	Type A + IEC 62955 RDC-DD allowed as alternative to Type B (per standards)
Built-in Type B in EVSE	Addendum does not require a duplicate board-mounted Type B if the EVSE incorporates one; the **final subcircuit** must still have compliant RCD protection on all live conductors per 7.9.3.3 and 2.6.3.3.1 (often Type A at the board minimum)

32 A socket for IC-CPD	Not banned domestically; **max demand assessment** and **load management recommended**
32 A plug-in leads	Should include warning about sustained 32 A on older installations
Equipment	IEC 61851 (EVSE), IEC 62955 (RDC-DD), IEC 62752 (IC-CPD)

Example — guideline vs Wiring Rules: Clause 7.9.3.3 already mandates Type B (or Type A + RDC-DD). The Addendum reinforces that residential circuits should originate from the **MEN board** and that a 32 A load is unlikely to be supported without assessing the rest of the house load — encouraging **smart load management** rather than assuming 32 A is always available.

6. Related Wiring Rules clauses

Clause	Application to residential 32 A EV
2.6.3.3.1 (NZ residential RCDs)	New/altered domestic final subcircuits need 30 mA RCD protection
2.6.2.2	Type AC RCDs not for new/altered work; use Type A, F, or B as applicable
2.2.2	Maximum demand — EV often counted at full rated current unless assessment/limitation/LMS used
2.3.3	Main switch at main switchboard controls whole installation (includes EV portion)
2.3.4.1 / 7.2.4.5	Outbuilding supplies — submain, MEN, testing if charger in detached garage board
3.4 / AS/NZS 3008	Cable size, derating, max 5% voltage drop (origin to load)
8.3.10 (ESR 2025 modification)	RCD testing must include DC leakage checks where applicable
Appendix P	Informative EV guidance (dedicated circuit, mechanical protection IP/IK)

7. Equipment and product compliance

- EVSE must have appropriate **Supplier Declaration of Conformity (SDoC)** under the ESR.
- Mode 3 wallbox: comply with **IEC 61851-1** (and 61851-22 where applicable).
- If using Type A + RDC-DD path: EVSE must include **IEC 62955** compliant RDC-DD (verify in SDoC).
- Outdoor install: typically **IP44 minimum**; IP55+ recommended for exposed weather.
- Impact protection: Appendix P suggests **IK07+** where vehicle damage is foreseeable.
- **EECA smart charger list** is voluntary (efficiency/OCPP) — not a substitute for Wiring Rules compliance.

8. Typical installation topology

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│  SERVICE / METER ──▶ MAIN SWITCHBOARD (MEN linked)            │
│        │                                                      │
│        ├── Existing house circuits (lights, power, HWS…)      │
│        │                                                      │
│        └── NEW: 32 A dedicated final subcircuit               │
│                  ├── 2-pole Type B RCBO (30 mA)  [or A+RDC-DD]│
│                  ├── Submain cable (sized per AS/NZS 3008)    │
│                  ├── 32 A isolator (adjacent to EVSE)         │
│                  └── Fixed Mode 3 EVSE (≥ 800 mm height)      │
│                            └──▶ Vehicle (Type 2 plug typical) │
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9. Worked example 1 — New garage wallbox (7.4 kW)

Scenario: 2024 house, 63 A single-phase supply, main switchboard in garage next to parking bay. Customer supplies 7.4 kW (32 A) Type 2 wallbox with integrated RDC-DD. Cable run 8 m from board to charger.

Design steps:

1. Confirm **maximum demand**: existing load + 32 A EV = often exceeds 63 A → apply **load management** or supply upgrade.
2. Install dedicated 32 A final subcircuit from MEN board — no other points on circuit.
3. Protect with **2-pole 32 A Type A RCBO (30 mA)** + EVSE RDC-DD per SDoC, OR **Type B RCBO**.
4. Cable: e.g. **6 mm²** copper (verify length, installation method, derating per AS/NZS 3008).
5. Mount EVSE ≥ **800 mm** AFFL; IP rating suitable for garage (moisture/dust).
6. **32 A isolator** within sight of EVSE (or use identifiable RCBO as isolation if visible).
7. Test: insulation, continuity, polarity, RCD trip, **DC leakage** per 8.3.10, EVSE earth monitoring.
8. Issue **CoC**; notify distributor if required.

Outcome: Compliant Mode 3 install under 7.9.3.3 with practical load management if oven + EV + heat pump coincide.

10. Worked example 2 — Existing home, limited capacity

Scenario: 1998 home, 50 A main switch, board full, no spare capacity for continuous 32 A. Customer wants overnight charging.

Options (in order of preference):

1. **Dynamic load management** (CT on mains, EVSE throttles to 6–16 A when house load high) — meets WorkSafe recommendation; may avoid supply upgrade.
2. **Limitation method** (per AS/NZS 3000 maximum-demand rules): documented design cap on total installation demand; user education on scheduling.

3. **Supply upgrade** (distributor): new mains, larger main switch, possibly three-phase.
4. Lower-rate charger (e.g. 16 A / 3.7 kW) on 20 A subcircuit if 7.9.3.2 Mode 2 socket route chosen — slower but smaller demand increment.

Important: Installing 32 A hardware without capacity assessment risks nuisance tripping, voltage drop, and overheating — non-compliant if installation becomes unsafe.

11. Worked example 3 — Detached garage subboard

Scenario: House MSB on dwelling; **submain** to garage distribution board 25 m away; charger mounted on garage wall.

Additional requirements:

- Confirm the charging final subcircuit originates from a **MEN switchboard** (WorkSafe Jan 2026 addendum). This is often the dwelling main switchboard; a detached garage distribution board only qualifies if the outbuilding supply and MEN arrangement comply with Clause 7.2.4.5 — confirm in design before issuing a CoC.
- Size **submain** for garage loads + 32 A EV (voltage drop over full path).
- Install EV protection (Type B or Type A + RDC-DD) at the **origin of the final subcircuit** — whichever board that is.
- Outbuilding MEN rules: polarity and MEN connection testing per Amendment 2 (8.3.9).
- Main switch still at dwelling MSB for whole installation (2.3.3).
- Local **32 A isolator** at EVSE strongly recommended (board not visible from charger).
- Mechanical protection: conduit/armour in driveway (WSX3 / Appendix H).

12. Protection options — Type B vs Type A + RDC-DD

Option	Board protection	EVSE requirement	Typical cost	Notes
A	Type B RCD/RCBO 30 mA, 2-pole	Standard Mode 3 EVSE	Higher (~$300–$500+ NZD)	Simplest compliance with 7.9.3.3(c)
B	Type A RCD/RCBO 30 mA, 2-pole	Built-in IEC 62955 RDC-DD	Lower board cost	Must verify RDC-DD in SDoC; common on quality wallboxes

Both options must disconnect **active and neutral**. Single-pole RCBOs are not suitable.

If the EVSE incorporates a Type B RCD, you may not need a second Type B at the board (WorkSafe addendum), but protection must still cover the **entire final subcircuit** from its origin to the EVSE — not only the charger enclosure.

13. Testing, certification, and records

1. Visual inspection: IP rating, height, labelling, dedicated circuit, isolator.
2. Tests per AS/NZS 3000 Section 8: continuity, insulation, polarity, earth fault-loop impedance where required.

3. **RCD operation test** — including **DC leakage** per modified 8.3.10 (2025 ESR). This verifies correct installation; it is not full product-type testing of an RDC-DD.
4. EVSE functional checks per manufacturer (e.g. earth continuity monitoring) where applicable.
5. Record date of energization at switchboard (2018 edition requirement).
6. Issue **Certificate of Compliance** to customer; retain test records.

14. Installer checklist

☐ Design to AS/NZS 3000:2018 + Amendments 1, 2, 3

☐ Clause 7.9.3.3: 32 A dedicated subcircuit, direct connect, 800 mm height, 32 A isolator

☐ Type B 30 mA (2-pole) OR Type A + IEC 62955 RDC-DD documented

☐ Charging final subcircuit originates from a MEN switchboard (WorkSafe addendum; confirm for detached garages)

☐ Maximum demand assessed; load management or upgrade if needed

☐ Cable sized per AS/NZS 3008; voltage drop ≤ 5%

☐ EVSE SDoC, IEC 61851, RDC-DD evidence if applicable

☐ Outdoor/mechanical protection as required

☐ Distributor notification if required by network rules

☐ Section 8 tests complete including RCD/DC leakage

☐ Certificate of Compliance issued

15. References and sources

1. Electricity Act 1992 — legislation.govt.nz
2. Electricity (Safety) Regulations 2010 — legislation.govt.nz
3. ESR Reg 59 — AS/NZS 3000 compliance — legislation.govt.nz
4. WorkSafe — Changes to ESR for electrical installations (Dec 2025) — worksafe.govt.nz
5. WorkSafe — EV charging safety guidelines — worksafe.govt.nz
6. WorkSafe — EV guidelines Addendum (Jan 2026) — worksafe.govt.nz
7. AS/NZS 3000:2018 — Standards Australia / Standards NZ
8. AS/NZS 3000 Amendment 1 — Clause 7.9 extract — electrotraining PDF
9. MBIE — Updating ESR standard citations — mbie.govt.nz
10. EWRB — Guidance on 2025 ESR changes — ewrb.govt.nz
11. Building CodeHub — AS/NZS 3000:2018 resource — building.govt.nz
12. Electric Vehicle Council (Australia) — supplementary EVSE installation guideline, not NZ law — electricvehiclecouncil.com.au

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